8. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG) S → A101A, A → 0A | 1A | ε

Aim:

To write a C program to check whether a given string belongs to the language defined by the CFG: $S \rightarrow A101A$ \(A \rightarrow 0A \mid 1A \mid \varepsilon\)

Algorithm:

1. Start the program.
2. Read a binary string from the user.
3. Find the length of the string.
4. Check whether the substring `101` occurs in the string.
5. If it occurs, display that the string belongs to the language; otherwise, display that it does not belong.
6. Stop the program.

Program :

main.c

```c
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i, len, found;

    for (int j = 1; j <= 2; j++)
    {
        printf("Enter binary string %d: ", j);
        scanf("%s", str);

        len = strlen(str);
        found = 0;

        for (i = 0; i <= len - 3; i++)
        {
            if (str[i] == '1' && str[i + 1] == '0' && str[i + 2] == '1')
            {
                found = 1;
                break;
            }
```

```
22                  break;
23              }
24          }
25
26          if (found)
27              printf("String belongs to the language.\n");
28          else
29              printf("String does not belong to the language.\n");
30      }
31
32      return 0;
33  }
34
```

Output :

```
Enter binary string 1: 0010
String does not belong to the language.
Enter binary string 2: 110110
String belongs to the language.
```

Result:

The C program was successfully executed to check whether the given strings belong to the language defined by the Context-Free Grammar (CFG). The first string was accepted, and the second string was rejected.